

Responsibility and Cognitive Power:

Exploring the hidden responsibilities behind artefacts within the **DESIGNER MAKER USER** exhibit at the **Design Museum**

Project Overview

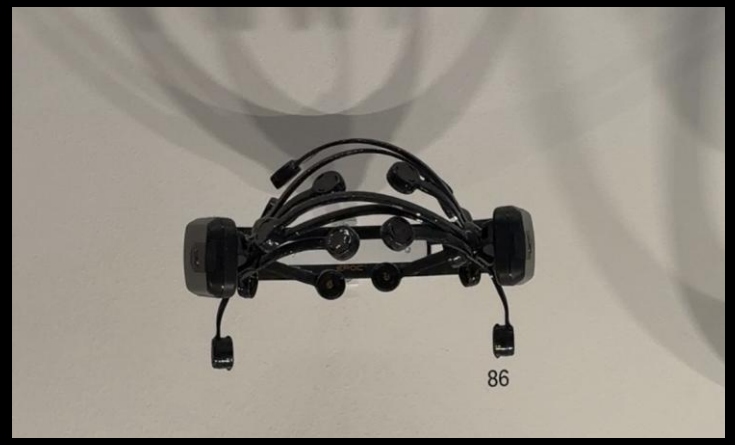
Within this unit, I am presented with core principles of responsible design. Exploring how environmental, social and ethical responsibilities operate within **contemporary design** practice. Looking at artefacts from the **DESIGNER MAKER USER** exhibit, I will examine how **responsibility is formed, shared, claimed or hidden** within design systems and these choices **affect people, communities and ecosystems**.

The core questions I will need to keep in my head will be; **how do design decisions shape human and ecological capability, and how is responsibility distributed, obscured or claimed by designers, makers and users within complex systems?** This project is not like my previous, there will be no solution because this project only focuses on the **first half of the double diamond (The defining/research aspect).**

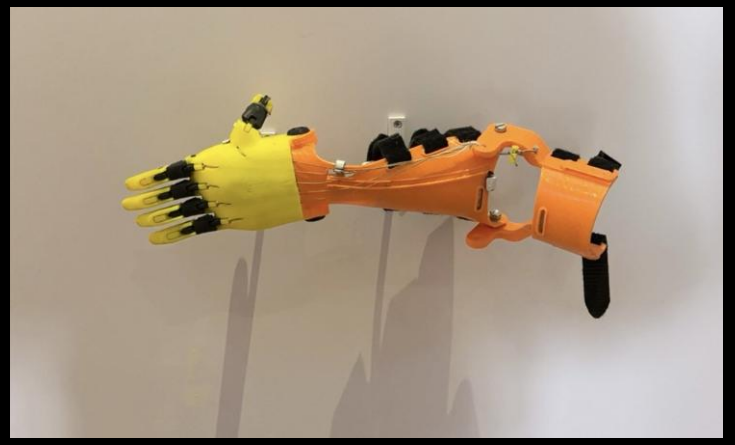
At the end of this, I should have produced:

- A **visual artefact** that makes responsibility in design visible.
- A **1,500-word analytical written reflection** that critically examines the same system.

Initial Artefact Selection



Emotiv EPOC Neuroheadset, 2007 Designed by Emotiv Systems and 4design.



e-NABLE Raptor Reloaded prosthetic hand, 2013-15 Designed by Ivan Owen, Andreas Bastian, Frankie Flood, Peter Binkley and Jeremy Simon.



iPod, 2001 Designed and manufactured by Apple.

Making of decision

Initial Artefact Selection

Before deciding on which artefact, I wanted to focus on, there were three that stood out to me the most in the exhibit. Which I displayed on the **previous slide**. All being completely different designs with zero correlation and in their function. However, were linked by one thing – their potential **hidden responsibility in design**.

To decide, I researched each artefact individually, asking critical questions linking to my project:

- The different **power structures** involved?
- Who gains and who loses **capability**?
- Where **responsibility** sits or becomes unclear?
- What **hidden costs** might exist?

Through said questions, I answered them for each artefact to help identify which had less or more depth to them for my project. Concluding, ruling out of iPod leaving me with the **prosthetic arm** and the **neuroheadset**. The iPod was strong in terms of:

- Lifecyle and environmental impact

BUT

- It was very similar to my last project

I wanted something to push me into new territory and choosing the iPod felt like a repeated task which did not appeal to me.

During week 3, I shared where I was at on **Padlet**. Explaining that these two artefacts stood out to me the most, I asked for **feedback from my tutor** and it was suggested that I should directly compare the two in terms of responsibility. Taking that approach, I created a quick test for myself asking simple but important questions like:

- Which one makes me more **curious**?
- Which one makes me slightly **uncomfortable** when I investigate it?
- Which one makes me want to **research more**?

Through this test, I had an epiphany realising that I was looking for something that created **friction** with me – something that was not only interesting but **challenging and unsettling** to think about.

Leading me to choose the neuroheadset over the prosthetic arm.

Chosen Artefact Analysis

What is it?

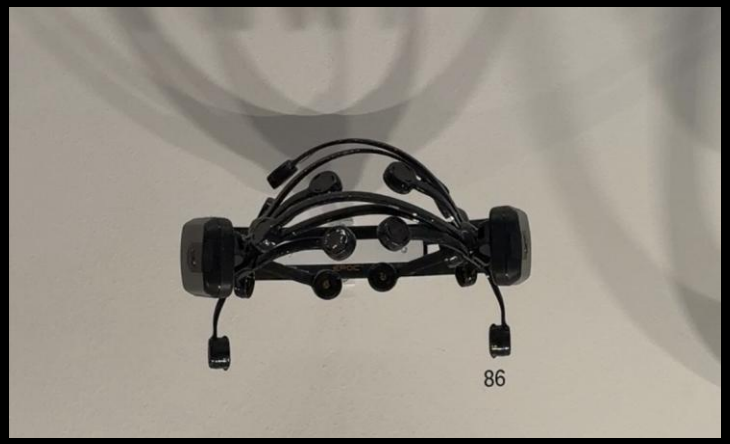
The **Emotiv EPOC Neuroheadset** is a **brain–computer interface (BCI) wireless headset** that allows users to **interact with digital systems using brain signals**. It detects **electrical activity in the brain** and translates it into commands that can **control computers, games, or other devices. (Emotiv.com. 2026 *Wearable EEG Sensor Headset: A Buyer's Guide)).**

What can it do?

The system can also **detect facial expressions, emotional states, and cognitive responses**, allowing **interaction beyond traditional input methods**. Users can **train the system** to recognise specific mental commands, creating a personalised interaction model based on their brain activity. Originally, **designed for the gaming industry, however, has expanded its audience including users with accessibility issues, researchers, developers and even research programs on driving a car**.

What kind of data is collected?

Looking at the **data and platform infrastructure**, it is stated that the EEG data that is collected consists of; **performance metrics, mental commands, affective data (meaning things to do with emotions) and eye and facial movement data**. These all used to aid in the function of the neuro-headset. Furthermore, this is done using the **electrodes Pg. 25 of the process book,** and they signal processing into the computer which is then processed as data and broken down. **(Emotiv.com. 2026, *Wearable EEG Sensor Headset: A Buyer's Guide).***



Emotiv EPOC Neuroheadset, 2007 Designed by Emotiv Systems and 4design.

Personal Thoughts

Chosen Artefact Analysis

How did I feel?

What drew me to the neuroheadset was how different it felt compared to everything else in the exhibition. It didn't just feel like a product — **it felt like a system with potentially serious implications.**

Straight away, I could see that something like this would come with a lot of hidden responsibility, especially because it deals directly with the brain. Exciting me because the product in general feels like a **real step into the future**. The idea of controlling systems through thought alone is powerful, especially for **accessibility and people with disabilities like;**

- Severe motor impairments
- Paralysis
- Communication disorders

But at the same time, it **raises concerns**.

Concerning myself with more critical themes like **data privacy, digital manipulation and cognitive liberty, linking directly** to the responsibility in design.

Once you start thinking about what's happening behind the scenes; **the data being collected, how it's used, and who controls it** — it becomes less straightforward than you may think. Feeling like something that could be beneficial, but also risky if not properly controlled or monitored by the right people.

Can be the future of interaction but also something dangerous if left unchecked.
This flooded my mind with multiple question like;

- **who owns** the data?
- who is being **monitored**?
- who **profits** from behavioural insight?
- what happens to **user autonomy**?
- **Is it safe** for a user's neurological system?

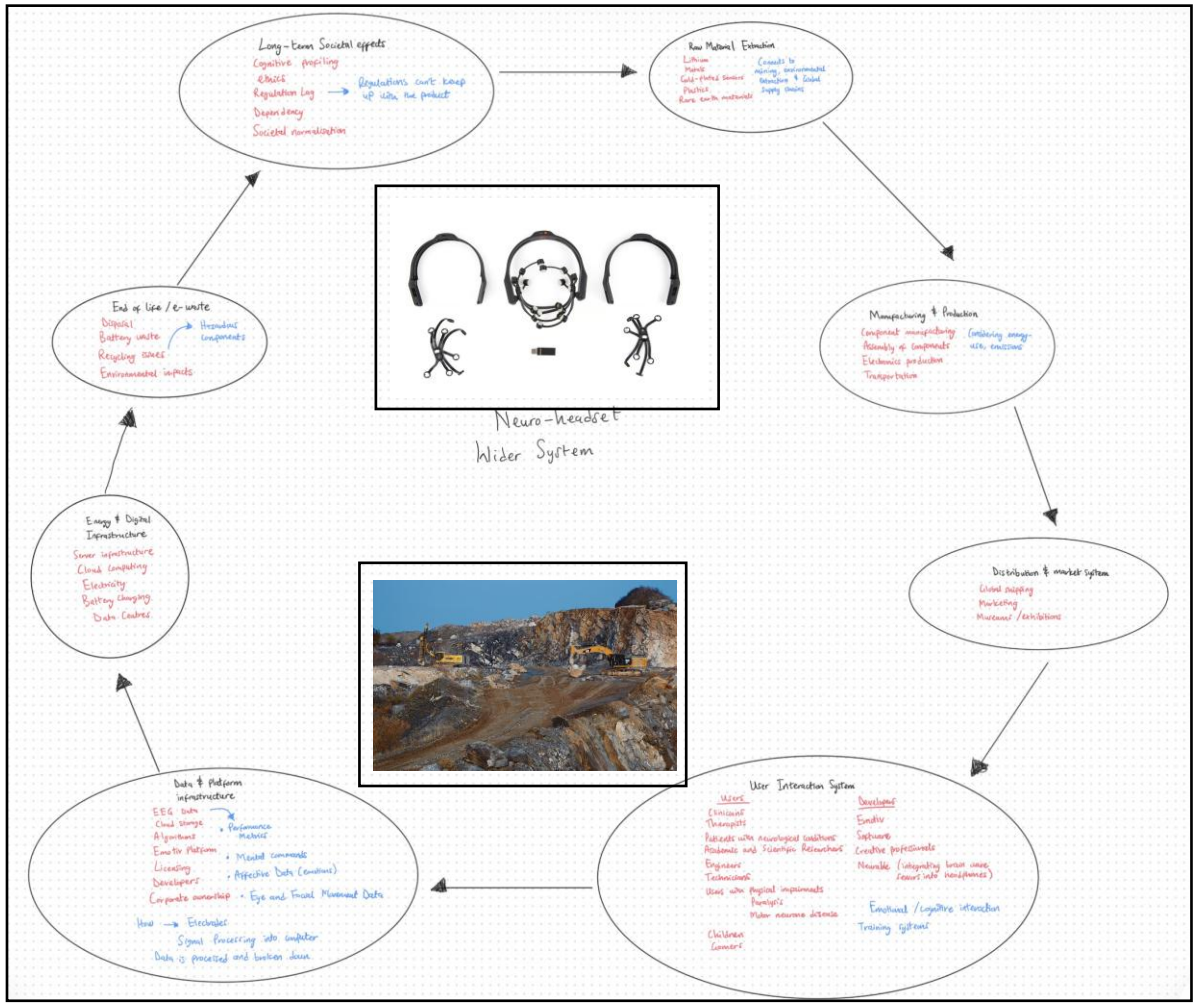
From Object→System

I realised this is not just an object.

What system does it sit within?

I immediately look at the **wider network, materials sourcing, maintenance requirements of parts and materials and the end-of-life cycle.** Through the **extraction of raw materials** there's manufacturing and material consumption which is **energy intensive, complex manufacturing and extraction of raw materials in harsh labour conditions** are used to source components in batteries like **lithium-ion.**

Additionally, components and parts like **pcb boards, rubber and plastic casing** are made with **durability and longevity in consideration**, **(Carbonclick (2025). *The environmental impact of electronics production and disposal*.)** therefore extending the functional lifespan of the device compared to other electronic devices. Felt pads are made to be made with a more sustainable sensor unit so that it doesn't corrode quickly, **reducing the need to replace and alleviating carbon footprint.**



Irwin, T. (2015) Transition Design
Life-Centred Design
"Design must consider impacts on all living systems, not only humans".

Ethics are not considered

Due to regulation lag
- Regulations/laws cannot keep up with rapid technological advancements, so they lag

Cognitive profiling

Increased dependency

Social normalisation

Long-term Societal effects

Raw earth materials
- Plastics (felt pads)
- Gold-plated sensors
- Lithium and other metals (for pcb boards)

Raw Material Mining/Extraction

This connects directly to, mining, intensive environmental extraction causing harm to the environment and global supply chains.



Considerations of energy use and emissions

Manufacturing and Production
- Component manufacturing
- Electronics production
- Assembly of components

Transportation

Recycling issues

Disposal issues
- Rubber and plastic casings

Battery waste
- Lithium-ion

Hazardous components

Environmental impacts/damage

End of life/E-waste

Emotiv EPOC Neuro-headset (Wider System)



Global shipping

Websites

Distribution and Market System
- Marketing
- Advertisements
- Museums
- Exhibitions

Server Infrastructure

Data Centres

Energy and Digital Infrastructure

Cloud computing
- Electricity usage
- Battery Charging

EEG Data
- Performance Metrics
- Mental Commands
- Affective Data (Emotions)
- Eye and Facial movement Data

Developers
- Emotiv Systems
- Emotiv Software platform
- Creative professionals
- Neurable (integrating brain wave sensors into head-set)

Data and Platform Infrastructure

How → Electrodes

Signal Processing into computer and then the data is processed and broken down
- Cloud Storage
- Algorithms
- Emotiv Platform
- Licensing
- Developers
- Corporate ownership

User Interaction System

Emotional/Cognitive Interaction

System has to be trained

Users
- Clinicians
- Therapists
- Academic and Scientific Researchers
- Engineers
- Technicians
- Children
- Gamers

- Patients with neurological conditions
- Physical impairments
- Paralysis
- Motor neurone disease

Where is Responsibility Claimed and Shared?

From Object → System

Claimed

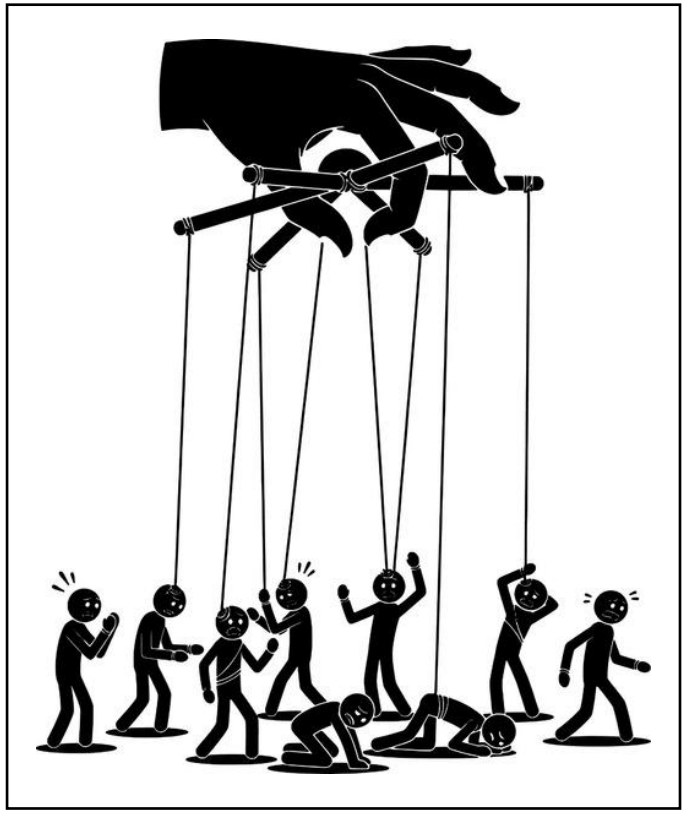
According to **Emotiv's terms of user data and terms of use**; Emotiv owns certain usages and other data generated or collected in connection with the use of their materials. Subsequently, they state that **users retain ownership of the scan data collected by the hardware that is uploaded to the service** however, you can only **access it through their software**. **(Emotivcloud.com. 2025, *Emotiv Term of Conditions & Privacy Policy*).**

This means that by using their product, **you grant the company a perpetual license and rights to use, reproduce, and analyse data**. Creating a situation where **ownership exists in theory, but control exists in practice**.

Shared

When it comes to who profits, this is shared between multiple companies; **Emotiv systems (primary owner of system + data), Developers building applications, Researchers using cognitive data and Designers like 4Design contributing to product development**.
(Emotivcloud.com. 2025, *Emotiv Term of Conditions & Privacy Policy*).

(This will be displayed throughout my process book, if unable to find, it will be on my responsibility map).



McDonough, W. and Braungart, M. (2002) *Cradle to Cradle Design: Remaking the Way We Make Things*
Externalities
"Costs are often hidden or displaced".

Where is Responsibility Hidden or Deferred?

From Object → System

Hidden

Initially, it needs to be understood **why responsibility matters so much** and that would be because of the **level of access this device has to our cognitive data**.
The idea that **"with access to brain data, the possibilities are vast" (Emotiv.com. 2026, *Wearable EEG Sensor Headset: A Buyer's Guide*.),** highlights both **opportunity and risk**. It may be able to empower users, but it also introduces concerns around:

- Data
- Privacy
- Surveillance
- Cognitive Liberty
- Potential Manipulation.

Additionally, the more advanced the headset **(higher channels)**, the **more detailed the data collected about a user's mental state**. Therefore, creating a situation where deeply personal information can be captured, often without the user fully understanding the extend of that access.

Deferred

Responsibility becomes very difficult to assign because **control, impact and accountability are distributed across different actors** and timeframes each holding **partial control but not fully accountable**.

(This will be displayed throughout my process book, if unable to find, it will be on my responsibility map).



Transition Design/Systems Design literature: Distributed responsibilities
"Responsibility is shared across systems".

What Happens to User Autonomy?

From Object → System

In some contexts, **autonomy is expanded but constrained in others**.
The headset can increase autonomy by **enabling control without physical movement and supporting users with disabilities like:**

- Physical impairments
- Paralysis
- Motor Neurone disease
- Communication disorders

However, it can also reduce autonomy because the headset "**requires training" (Emotivcloud.com. 2025, *Emotiv Term of Conditions & Privacy Policy*). which means you can not rely on the product without sufficient training**.
Additionally, it can also **misinterpret brain signals** which causes frustration. All these negatives **reduces the trust** from users because of the headset's **limited reliability**.

Boylston, S. (2019), Chapter 8.
Access ≠ Capability
"Access does not guarantee opportunity".

Boylston, S. (2019), Designing with Society, Introduction
Capability/Design Purpose
"The capabilities approach focuses on what people are able to do and be".

Capability

From Object → System

Who gains?

- Patients with neurological conditions
- Physically impaired users
- Users with a form of paralysis preventing them from tactile interactions.
- Users with Motor neurone disease.
- Gamers
- Clinicians
- Researchers with access to cognitive data.
- Companies with innovation and data advantage.

Who loses?

- Every user I listed in the **gain** section lose cognitive privacy.
- Users also lose full autonomy due to system control and interpretation errors.
- Users that may lack knowledge on the device like children.
- Patients that need to be in constant care.
- The wider society in general at risk of potential normalisation of brain data extraction.

Who is excluded?

- People who can’t afford the device.
- People without technical understanding.
- People outside research environments.

Access

From Object → System

Who participate?

- Researchers
- Emotiv
- Software developers
- Creative professionals
- Neurable (adding brain waves signals into headphones)
- Clinicians
- Care takers of in able patients
- Gamers
- Tech-literate users
- Some disabled users (depending on context)

Who can not?

- General public due to the cost and complexity, also may not be trust.
- Non-technical users.
- Older people that may not understand the system, they will not be able to train them or find it difficult to.
- Children due to parental supervision.
- Users that need caretakers.
- Patients that need Nurses or Doctors to help them due to their physical impairments or paralysis.
- People without institutional access.

Hidden barrier

Just because you **have access**, does not mean you fully understand the system, able to control it or your data. The headset may be able to increase user autonomy, but it can also reduce it. This is because, it requires training and even with a mastered level of training, the headset can misinterpret signals sent by your brain which makes it less reliable.

Referencing one of the readings (look below). It states; “Access does not guarantee opportunity”, depicting that even if users have the headset, they may not control or benefit from their own brain data.

Boylston, S. (2019), Chapter 8.
Access ≠ Capability
“Access does not guarantee opportunity”.

Power

From Object → System

Who decides?

- Emotiv systems because they are the primary owners and controls the platform and data.
- Designers shape interaction whilst developers shape applications.
- Manufacturers and distributers
- Engineers of the internal component and external structure of device.
- Technicians
- 4Design deciding on what kind of design they want for the product.
- Researchers that gain cognitive data from users.

Who benefits?

- Emotiv benefits because of data and commercialisation.
- Designers like 4Design contributing to product development.
- Developers building the applications.
- Software developers
- Creative professionals
- Researchers gain data insights which benefit them.
- Developers, from applications.
- Patients with neurological conditions and physical impairments.

Who bears the cost?

- Every user that can access and use the product due to privacy risk and data exploitation.
- Society bears the costs of long-term implications when it comes to brain data systems.
- Countries where raw mining and extraction of materials occur.
- The environment after disposal of hazardous waste.

Insights

From Object → System

Capability

The neuroheadset expands functional capability and as it is doing so, it simultaneously introduces new forms of dependency **(gains)** and vulnerability **(loses).**

Access

Access to the system does not equate to meaningful participation, because users have limited control over how their cognitive data is used.

Power

Power is concentrated within the systems infrastructure, while users generate most of the value without having equivalent control.

Boylston, S. (2019), Chapter 8.
Access ≠ Capability
“Access does not guarantee opportunity”.

Boylston, S. (2019), Designing with Society, Introduction
Capability/Design Purpose
“The capabilities approach focuses on what people are able to do and be”.

McDonough, W. and Braungart, M. (2002) *Cradle to Cradle Design: Remaking the Way We Make Things*
Externalities
“Costs are often hidden or displaced”.

Why is Responsibility Important? (through a power lens)

From Object → System

Using a **power lens** exposes aspects of the neuroheadset that are not explicitly visible when you focus on only functionality or innovation. The device appears to **empower users by enabling interaction** through brain signals sent by the users however, a deeper analysis shows that **control over the system is not equally distributed**.

Companies such as **Emotiv retain significant power** through their **control of data infrastructure**, while **users generate the valuable cognitive data with the cost of not having full access or control over how it is used and what it is used for**. Highlighting a disconnection between **perceived empowerment and actual authority** within the system.

Thinking as an emerging UX designer, I feel challenged to think beyond usability and to consider **how design decisions shape power dynamics**. In return raising important questions about **who benefits from interaction systems and who is responsible for protecting users**, as a matter of fact, whether they are **protected** or not in the first place, especially when it comes to dealing with **sensitive and unsecure data** like brain activity.

McDonough, W. and Braungart, M. (2002) ***Cradle to Cradle Design: Remaking the Way We Make Things***
Externalities
“Costs are often hidden or displaced”.

Transition Design/Systems Design literature:
Distributed responsibilities
“Responsibility is shared across systems”.

Boylston, S. (2019), Chapter 8.
Access ≠ Capability
“Access does not guarantee opportunity”.

Enquiry Draft Questions

Enquiry Development

These are the questions I asked myself and topics I considered as I planned my enquiry statement and draft:

What system I am analysing?
The product being the **neuroheadset** and its system whilst implying boundaries of brain data, interaction system and data infrastructure.

Why responsibility matters?
I will need to highlight the high stakes when it comes to **cognitive data of users, ethics in privacy and autonomy then finally power** (who controls data).

Ethical tensions/trade-offs?
- Showing that understand where there is tension:
- Empowerment vs Control
- Capability vs Exploitation
- Innovation vs Regulation



Enquiry Draft Statement

This project explores how responsibility operates with the Emotiv EPOC neuroheadset system. It focuses on design decisions shaping cognitive capability, access and power for users and others. It works by examining how brain data is collected and used however, responsibility becomes unclear, contested and even displaced between designers, companies and regulatory bodies.

Final Enquiry Statement

Enquiry Development

This project examines how responsibility operates within the **Emotiv EPOC Neuroheadset system**, focusing on how design decisions can affect and shape **cognitive capability, access and power for users, researchers, technology companies** and all affected actors within the system. It investigates how the collection and use of brain data reveals **ethical tensions** between **technological empowerment and cognitive autonomy** and where responsibility for cognitive data is **obscured, unevenly distributed, or deferred across designers, corporations and regulatory systems.**

Final Enquiry Question

How do design decisions within the **Emotiv EPOC Neuroheadset system shape cognitive capability and autonomy**, and how is **responsibility** for brain data **obscured, distributed, or deferred across actors** that **gain** based on **capability**?

Boylston (2019)

Theoretical Framework/Readings

Capability / Design Purpose
Quote
“The capabilities approach focuses on what people are able to do and be.”

Statement
Design should expand people’s real opportunities, not just provide functional solutions.

Insight to Project
The neuro-headset doesn’t just *enable interaction;* it shapes what users can do cognitively. This doesn’t automatically mean autonomy or empowerment.

Reference
Boylston, S. (2019) Designing with *Society*, Introduction.

Access ≠ Capability
Quote
“Access alone does not guarantee opportunity.”

Statement
Providing access to a system does not ensure users can benefit from it.

Insight to Project
Even if users have the headset, they may not be able to control or even benefit from their own brain data.

Reference
Boylston, S. (2019) Chapter 8.

Systems Thinking
Quote
“Contemporary challenges are interconnected and cannot be addressed in isolation.”

Statement
Design operates within complex, interconnected systems where changes affect multiple outcomes.

Insight to Project
The neuro-headset device isn’t standalone but sits inside a data + cognitive + platform ecosystem.

Reference
Boylston, S. (2019) Section 1 / Chapter 1.

Boylston (2019)

Theoretical Framework/Readings

Long-Term Thinking / Antifragility
Quote
“Short-term thinking undermines long-term resilience.”

Statement
Design must consider long-term systemic impact rather than immediate outcomes.

Insight to Project
Cognitive risks from the headset may emerge later, not during initial/short-term use.

Reference
Boylston, S. (2019) Chapter 6.

Pace Layers
Quote
“Civilisation operates across layers that change at different speeds.”

Statement
Fast-moving technologies often outpace slower systems like culture and regulation.

Insight to Project
Neurotech evolves faster than ethics + law that lag → responsibility gap

Reference
Boylston, S. (2019) Chapter 6.

McDonough & Braungart (2002/2007)

Theoretical Framework/Readings

Less Bad ≠ Good
Quote (direct from talk)
“Being less bad is not being good.”

Statement
Reducing harm is insufficient; design should create positive impact.

Insight to Project
Even if the product is “ethical,” it may still be extractive by design.

Reference
McDonough, W. and Braungart, M. (2002/2007) *Cradle to Cradle Design: Remaking the Way We Make Things and* TED Talk.

Lifecycle Thinking
Quote
“Design should consider the full lifecycle of products.”

Statement
Design must account for impacts beyond the point of use.

Insight to Project
Brain data doesn’t stop at interaction — it persists, gets stored, reused, sold.

Reference
McDonough, W. and Braungart, M. (2002) *Cradle to Cradle Design: Remaking the Way We Make Things*

McDonough & Braungart (2002/2007)

Theoretical Framework/Readings

Externalities
Quote
“Costs are often hidden or displaced.”

Statement
Design systems frequently shift consequences onto users or society.

Insight to Project
Users may carry long-term cognitive/privacy risks, not the company.

Reference
McDonough, W. and Braungart, M. (2002) *Cradle to Cradle Design: Remaking the Way We Make Things*

Linear vs Cyclical
Quote (direct concept)
“Waste equals food.”

Statement
Systems should be regenerative rather than linear.

Insight to Project
Current neurotech is extractive (data out, nothing returned).

Reference
McDonough, W. and Braungart, M. (2002) *Cradle to Cradle Design: Remaking the Way We Make Things*

Lutz/Frameworks

Theoretical Framework/Readings

Life-Centred Design
Quote
“Design must consider impacts on all living systems, not only humans.”

Statement
Design should account for ecological and societal systems beyond the user.

Insight to Project
The neuro-headset affects society + cognition + data ecosystems, not just individuals.

Reference
Irwin, T. (2015) *Transition Design*.

Interconnection
Quote
“Systems are interconnected and interdependent.”

Statement
Design decisions propagate across systems.

Insight to Project
Brain data feeds larger AI/data infrastructures

Reference
Irwin, T. (2015).

Distributed Responsibility
Quote
“Responsibility is shared across systems.”

Statement
No single actor fully controls outcomes in complex systems.

Insight to Project
No one is fully accountable for misuse of brain data → accountability gap

Reference
Transition Design / Systems Design literature

Secondary Research

Through secondary research, I was able to answer early questions that were important due to the subjects surrounding my chosen artefact and develop a deeper understanding of the **wider system it operates within**.

At the beginning of the project, I had little to no prior knowledge of the **Emotiv EPOC Neuroheadset** or the **ethical and social responsibilities** connected to **brain–computer interaction** technology. Researching **articles, company information, academic texts, and discussions** around cognitive data allowed me to move beyond the analysis of the object alone and begin to focus on investigating the **hidden systems, responsibilities, and power structures** connected to it.

The information gathered through secondary research became the **foundation** for the rest of my project, influencing my **enquiry question, systems maps, critical arguments, and final artefact** direction. The following pages will document the **key questions** explored through research and the **insights** that informed the development of my project.

Emotiv Wearable EEG Guide

Secondary Research

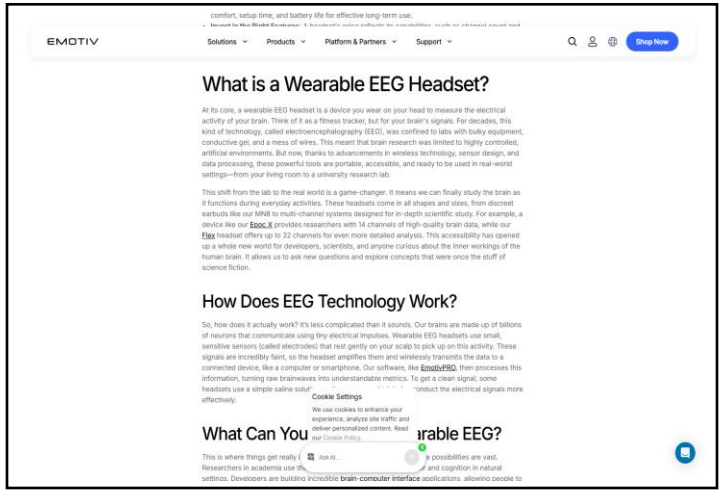
Emotiv Wearable EEG Guide
Emotiv.com. (2026). *Wearable EEG Sensor Headset: A Buyer's Guide*. [online] Available at: https://www.emotiv.com/blogs/news/wearable-eeg-sensor-headset-guide [Accessed 10 May 2026].

What I learned?
This source aided me when it came to understanding how EEG neuro-headsets function. Additionally, how they function and how the Emotiv EPOC translates electrical brain activity into digital interaction for users. The source also explained how the system can capture cognitive signals communicated by the users internally and converts them into commands for external use within systems.

This introduced the idea that brain activity itself becomes usable data.

Why was it important?
This was important because it helped me move beyond seeing the headset as a physical object displayed in a museum and instead understand it with more depth – as part of a larger cognitive data system.

Key takeaway
The neuro-headset does not only enable interaction – it has a deeper system that communicates to transform brain activity into useable data.



McDonough, W. and Braungart, M. (2002)
Cradle to Cradle Design: Remaking the Way We Make Things
Lifecycle Thinking
"Design should consider the full lifecycle of products".

Springer Academic Article

Secondary Research

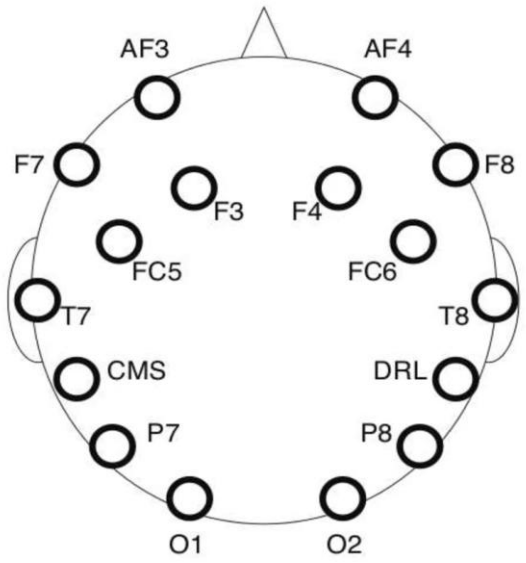
Springer Academic Article
Duvinage, M., Castermans, T., Petieau, M., Hoellinger, T., Cheron, G. and Dutoit, T. (2013). Performance of the Emotiv Epoc headset for P300-based applications. *BioMedical Engineering OnLine*, 12(1), p.56. doi:https://doi.org/10.1186/1475-925x-12-56.

What I learned?
This paper provided me when technical insight into the structure and the capabilities of the Emotiv EPOC system. This including things to do with its sensors, electrodes and ability to perform brain-computer interaction tasks.

Therefore, demonstrating how the headset has applications not just for entertainment but research and accessibility support.

Why was it important?
This was important because the source was able to reinforce the idea of the headset expanding capability and accessibility as it increases the amount of cognitive data collected from users.

Key takeaway
Increased capability requires increased cognitive data extraction.



Emotiv electrodes
The headset is composed of 14 different electrodes which when the number is increased; brain data is scanned in more detail.

Emotiv Terms & Conditions/Privacy

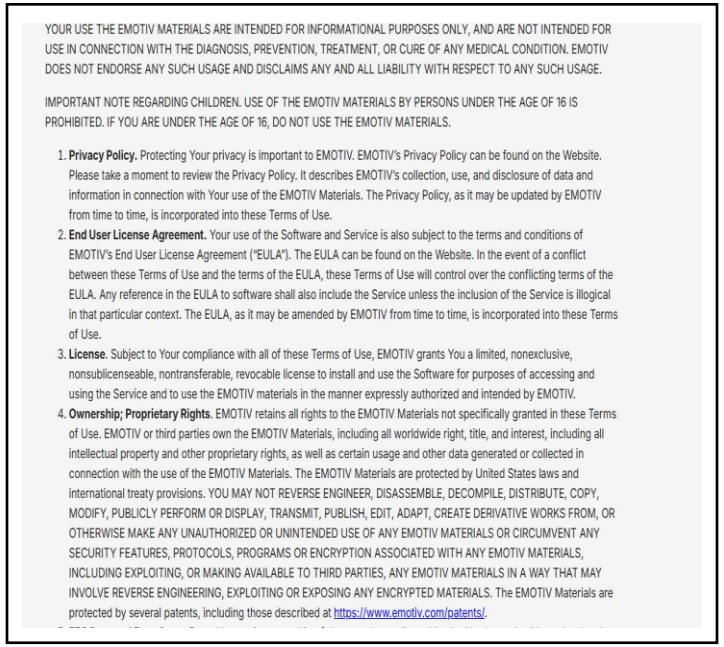
Secondary Research

Emotiv Terms & Conditions/Privacy
Emotivcloud.com. (2025). *Emotiv Term of Conditions & Privacy Policy*. [online] Available at: https://id.emotivcloud.com/eoidc/privacy/terms/ [Accessed 10 May 2026].

What I learned?
This source revealed to me how Emotiv manages their ownership, licensing and access to user cognitive data. It is stated that although, users retain ownership of their data, the company is granted extensive rights to analyse, use and reproduce any information they receive through their platform.

Why was it important?
This was important because the exposed the imbalance between theoretical ownership and practical control; providing value and still having less power. Supporting my investigation into responsibility, autonomy and power.

Key takeaway
Ownership exists in theory, but control exists in practice.



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Boylston, S. (2019), Chapter 8.
Access ≠ Capability
"Access does not guarantee opportunity".

Environmental Impacts of Electronics

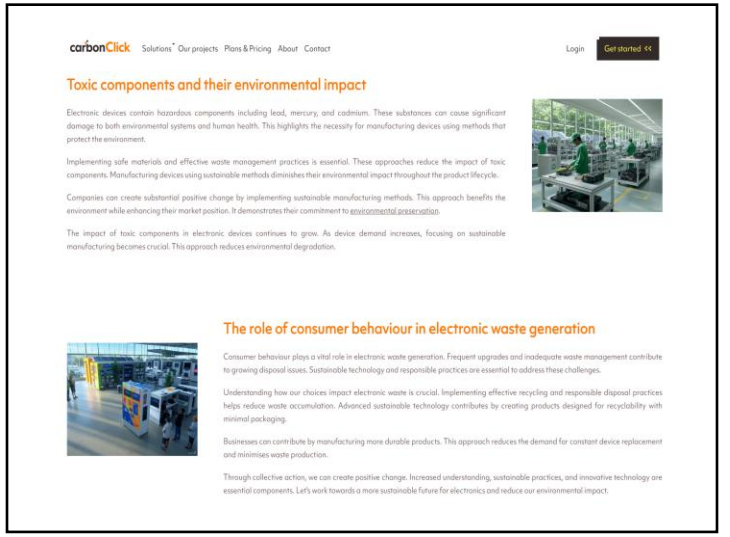
Secondary Research

Environmental Impact of Electronics
Carbonclick (2025). *The environmental impact of electronics production and disposal*. [online] The environmental impact of electronics production and disposal. Available at: https://www.carbonclick.com/news-views/the-environmental-impact-of-electronics-production-and-disposal.

What I learned?
This source helped me connect the environmental consequences to electronic products like the neuro-headset. Involving resource extraction, manufacturing emissions, e-waste and disposal issues.

Why was it important?
This was important because my understanding of responsibility beyond the user experience has expanded, allowing me to see beyond users and look at hidden environmental costs across time and geography.

Key takeaway
The impacts of technology extend beyond interaction within wider environmental systems.



McDonough, W. and Braungart, M. (2002/2007) *Cradle to Cradle Design: Remaking the Way We Make Things and* TED Talk.
Less Bad ≠ Good
“Being less bad is not being good”.

Irwin, T. (2015) Transition Design
Life-Centred Design
“Design must consider impacts on all living systems, not only humans”.

Powerhouse Museum Collection Entry

Secondary Research

Powerhouse Museum Collection Entry
Powerhouse.com.au. (2023). *Powerhouse Collection - 'Emotiv EPOC' brain-computer interface wireless headset*. [online] Available at: https://collection.powerhouse.com.au/object/416608 [Accessed 10 May 2026].

What I learned?
I was provided historical and curatorial context surrounding the Emotiv EPOC headset by this source, including its intended purpose, cultural significance and positioning as a rapidly emerging interaction technology.

Why was it important?
This was important because it helped situate the neuro-headset within a broader design give it great technological context instead of projecting it only as a consumer product.

Key takeaway
The neuro-headset represents a shift toward increasingly immersive forms of interaction.



4Design Project Page

Secondary Research

4Design Project Page
4design. (2019). *EPOC Headset by Emotiv | Industrial Design | 4DESIGN*. [online] Available at: https://www.4design.com.au/project/epoc-eeg-headset/ [Accessed 10 May 2026].

What I learned?
This source explained the design intentions that were behind the neuro-headset; this included usability, ergonomics, accessibility and interaction goals.

This demonstrated how the product was intentionally designed to feel futuristic, intuitive and empowering.

Why was it important?
This was important because it supported me in the understanding of how design contributes to shaping perceptions of innovation, trust and empowerment.

Key takeaway
Design decisions are what influence how technological systems are perceived are accepted.



A headset that reads your brainwaves | *Tan Le* Youtu.be. (2026). Available at: https://youtu.be/fVhggGSjXVg?si=FgC2jpWD7tgQ1XQA [Accessed 10 May 2026].

Emerging Insights

Secondary Research

- **“Capability expansion depends on cognitive data extraction”.**
- **“Responsibility becomes distributed across invisible systems”.**
- **“Access does not guarantee autonomy”.**

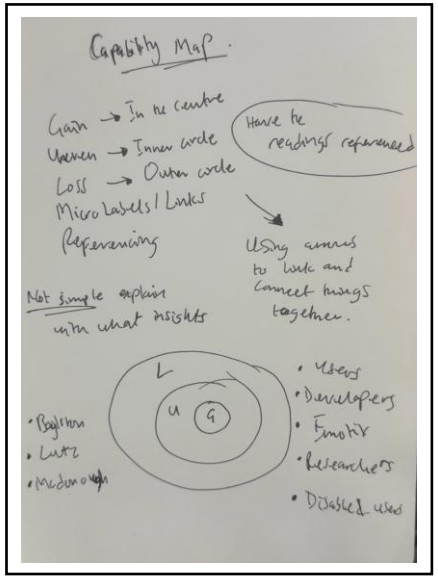
Boylston, S. (2019), Chapter 8.
Access ≠ Capability
“Access does not guarantee opportunity”.

Transition Design/Systems Design literature:
Distributed responsibilities
“Responsibility is shared across systems”.

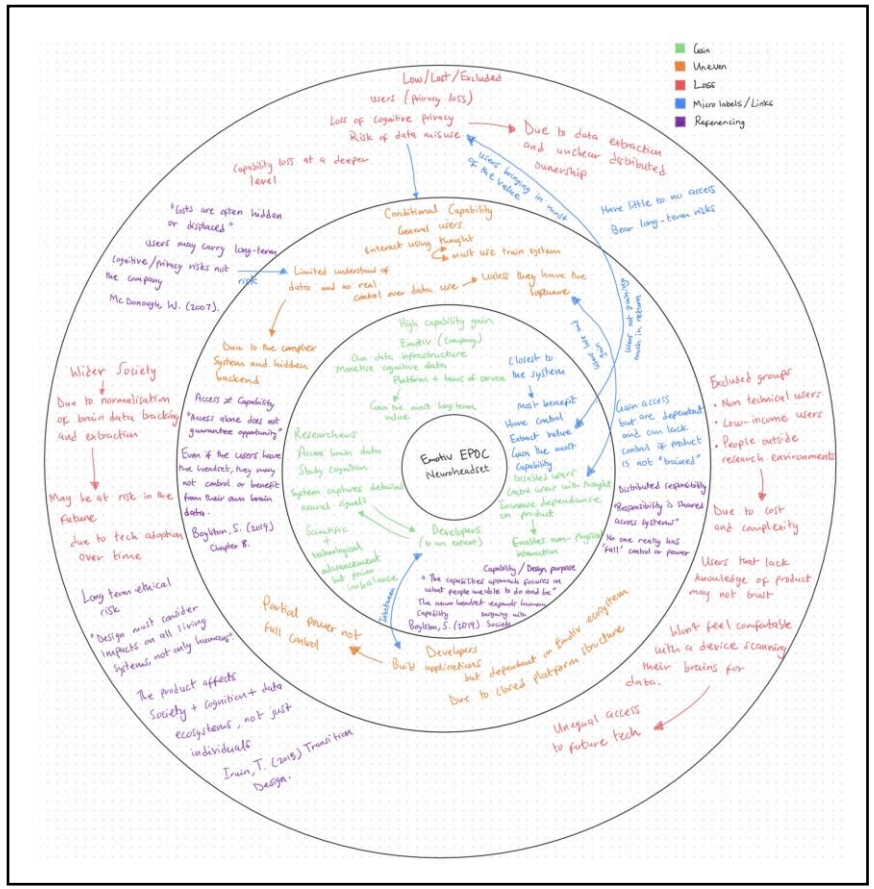
McDonough, W. (2007)
Linear vs Cyclical
“Waste equals food”.

Capability Map

Mapping Development



- **Initial planning** and brainstorming before I begun my draft.



- **My mapping draft**, including everything consisting of capability within the neuro-headset's system. I created a finalised map which will be presented on the next slide due to feedback from my peers to make this digitally for legibility.

Insight

The neuroheadset redistributes capability unevenly – it expands the ability of functionality for some users while introducing new dependencies, exclusions and loses of cognitive autonomy for others

Finalised Capability Map

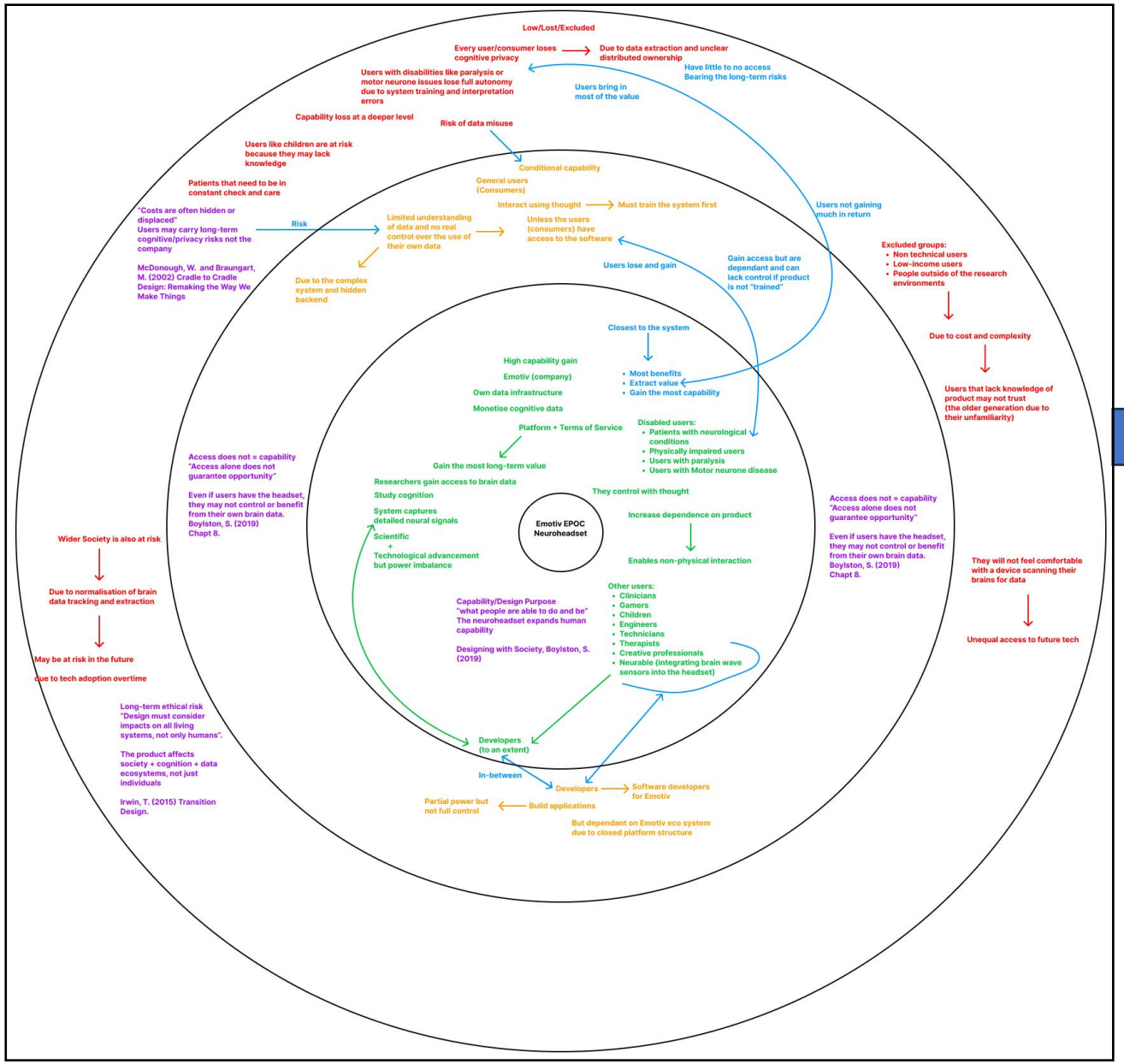
Mapping Development

Map Caption

This map displays the system of capability within the neuroheadset visually showing how the **Emotiv EPOC neuroheadset redistributes cognitive capability across different actors within the system.** As stated in my insight; the device **expands functionality capability for users**, particularly those with **physical impairments** to aid them in using devices without having to worry about obstacles. However, this **capability is uneven and conditional**.

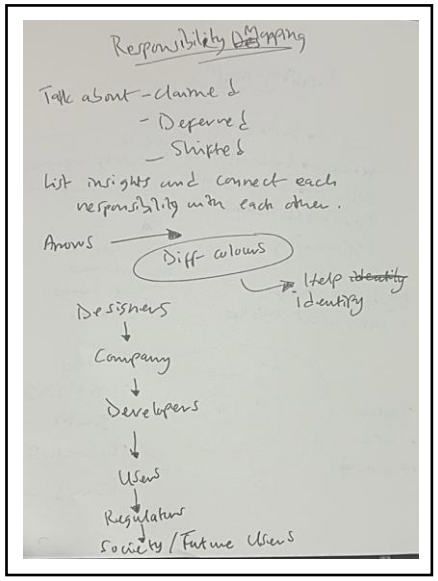
Access to the system does not guarantee full autonomy, due to users lacking control over how their brain data is captured and used. At the same time, companies and researchers gain significant capability through access to cognitive data, highlighting an **imbalance in power and value distribution** even though most of the incoming value are from the users.

Drawing on **Boylston's capabilities approach**, the map exposes that capability is not simply about access to technology, but about the **ability to meaningfully act within and control the system**. Therefore, revealing a **shift from empowerment at the surface level to dependency and vulnerability at a much deeper systemic level**.

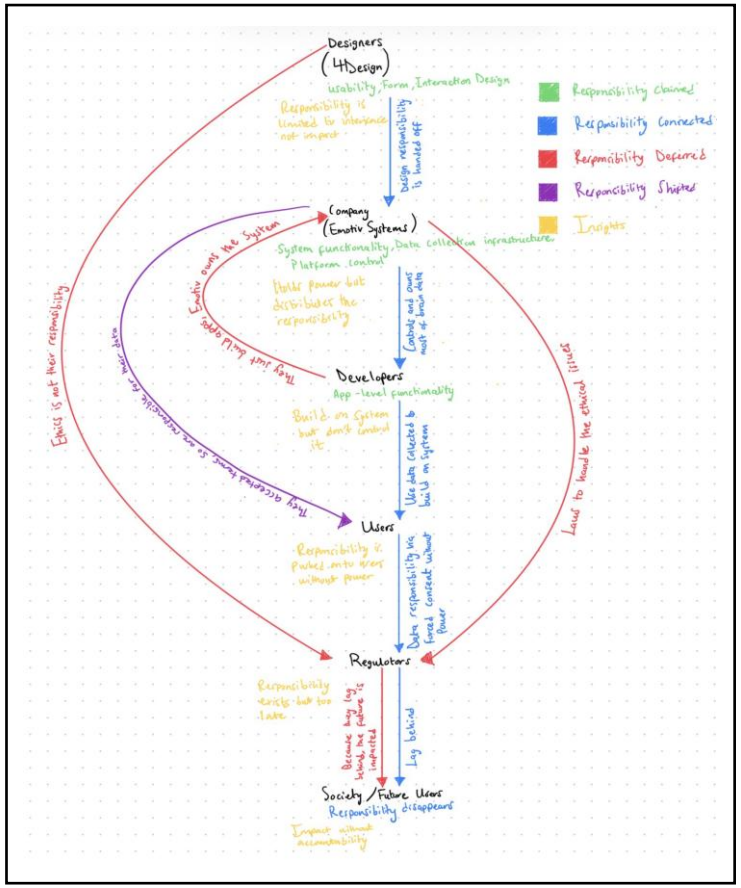


Responsibility Map

Mapping Development



- **Initial planning** (same explanation on slide 32)



- **My mapping draft** (same explanation on slide 32)

Insight

Responsibility is distributed unevenly. Strategically shifted and deferred away from companies with power towards those with least power and control. In return, long-term consequences are unaccounted for.

Finalised Responsibility Map

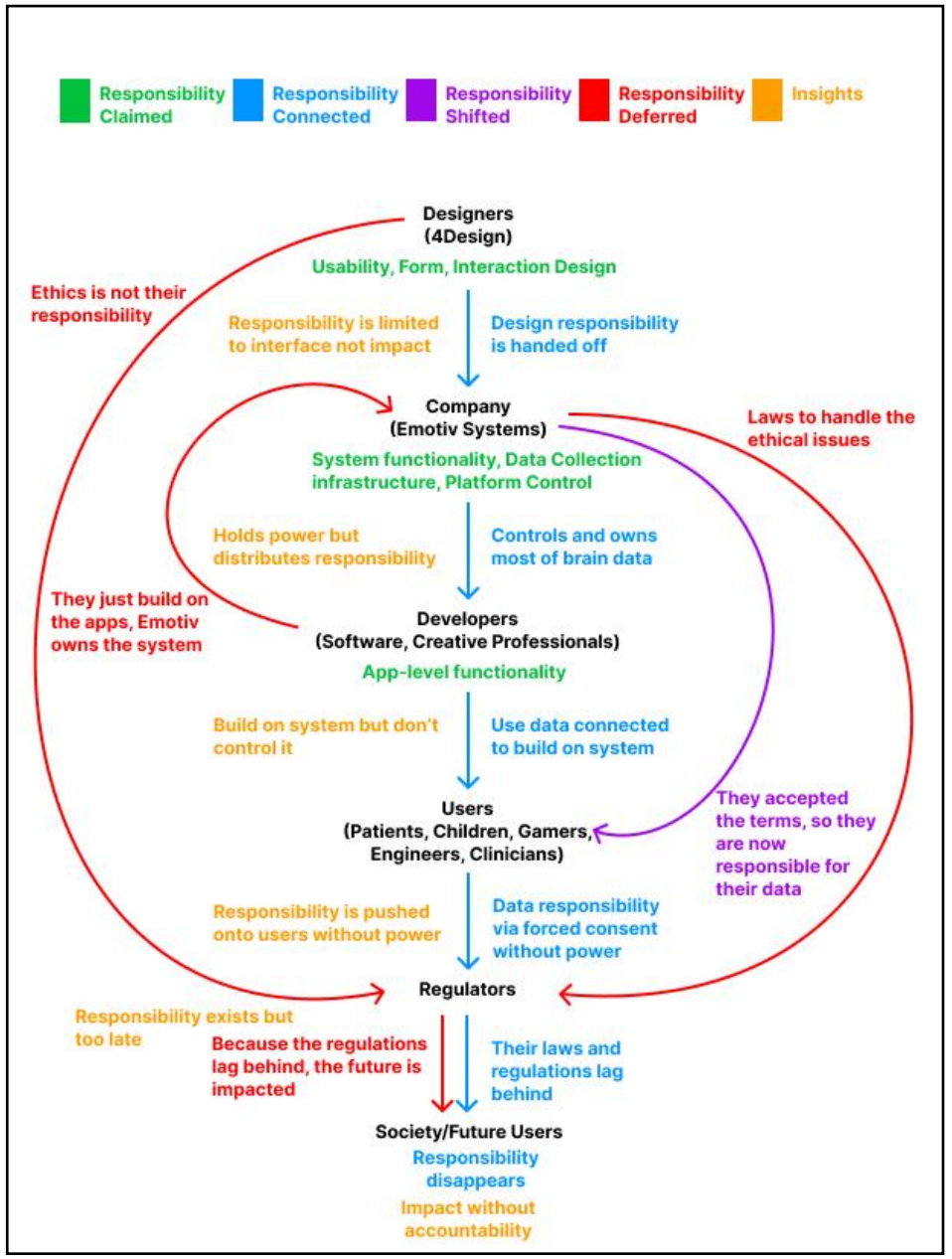
Mapping Development

Map Caption

This map reveals how responsibility within the **Emotiv EPOC neuroheadset system is distributed, deferred, shifted across various actors.** Various designers and developers claim responsibility for functionality and usability; this is because it **benefits their company and reputation**. However, when it comes to the **responsibility for the ethical implications of their brain data collection, it is frequently deferred to companies, users or regulatory systems with less power**.

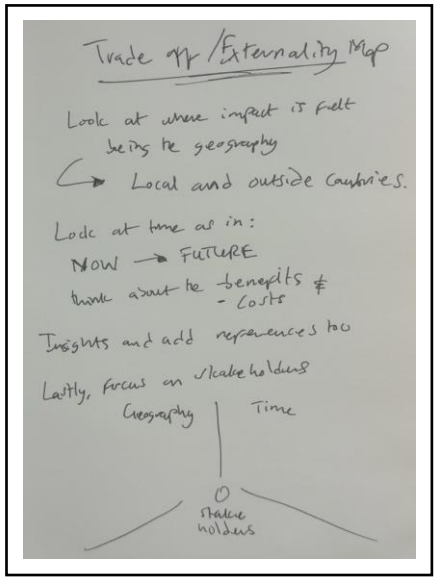
Emotiv Systems keep a significant amount of **control over data infrastructure yet shifts responsibility onto users through consent-based agreements that conceal the complexity of data ownership and use.** At the same time**, regulators** struggle to keep pace with the **rapid technological advancements**. Therefore, resulting in **gaps where accountability should be taken but was unclear or absent**.

Ultimately, responsibility for **long-term societal impacts**, specifically regarding cognitive privacy, disappears entirely and leaves future users and their wider society to **bear the consequences with no control over it** since every cause of action would be behind them in time. **Highlighting how responsibility within the system is not only distributed but strategically displaced**.

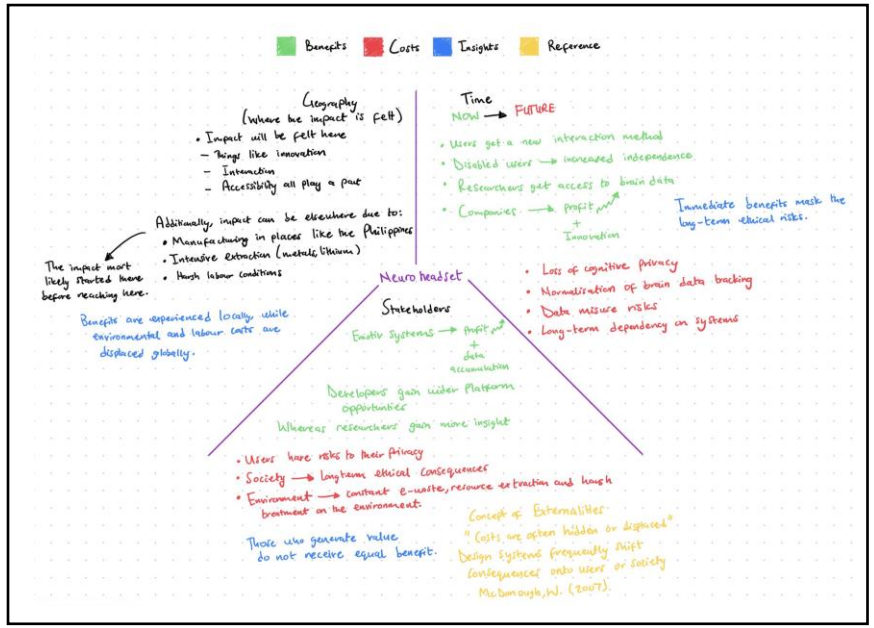


Trade Off/Externality Map

Mapping Development



- **Initial planning** (same explanation on slide 32)



- **My mapping draft** (same explanation on slide 32)

Insight

The neuroheadset creates value in the present. As doing so, it also displaces the social, ethical and environment costs across time, geography and the less powerful groups.

Finalised Trade Off/Externality Map

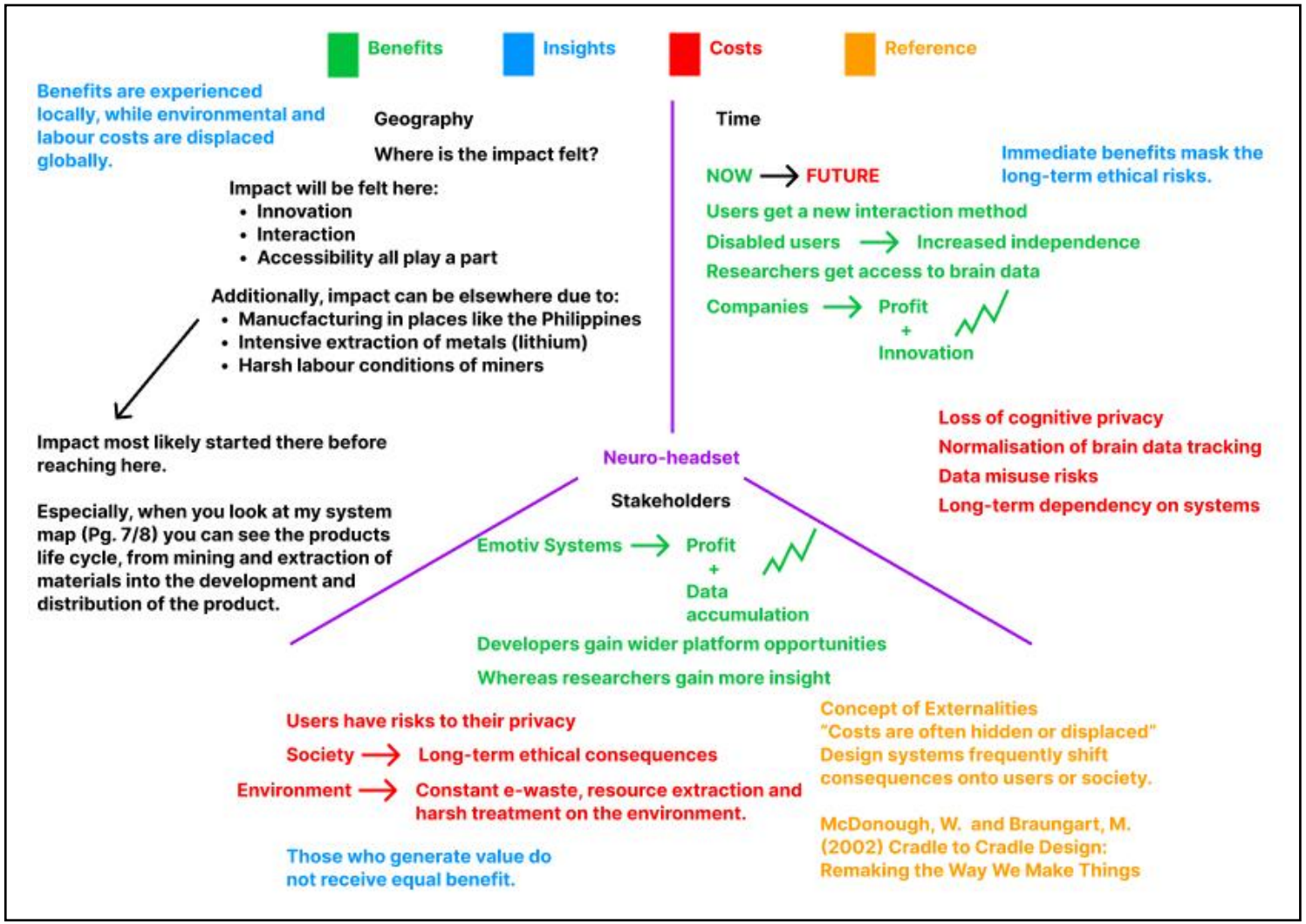
Mapping Development

Map Caption

This map visualises how the **Emotiv EPOC neuroheadset displaces its impacts across time, geography and stakeholders.** This system delivers various positives like **enhanced interaction, accessibility and data-driven innovation**; however, these advantages are unevenly distributed. The **long-term consequences like; risks to a user's cognitive privacy, misuse of their data and even societal dependence** on the headset are deferred to future users and its broader society.

At the same time, the **environmental and material costs of production**, including resource extraction and electronic waste are geographically displaced to regions **outside the primary user location**.

This draws onto **McDonough and Braungart's concept of externalities; the map revealing how "costs are often hidden or displaced"** and exposing how the system operates within a linear model that generates value in the present, **while shifting responsibility and harm elsewhere**. Highlighting the **ethical imbalance** at the core of the system.



Capability is Conditional

Critical Insights/Emerging Arguements

Emerging Argument
The neuro-headset expands functional capability. It does this by enabling new forms of interaction and accessibility through brain-computer interaction. However, these expanded capabilities are conditional and that is because participation within the system is greatly dependant on the continuous extraction and process of cognitive data. The cycle of how the system functions, relies on each other to work.

As some users gain increased interaction and accessibility like; users with severe paralysis being able to interact with technological products, others also experience new forms of dependency, vulnerability, exclusion and reduced cognitive autonomy.

Key Insight
For capability to be expanded, cognitive data needs to be extraction.

Why This Matters
This challenged me as a designer to think beyond the usability a product/device and recognise that not only does it provide functionality, but it also shapes what users are able to do and become.

Boylston, S. (2019), Designing with Society, Introduction Capability/Design Purpose
“The capabilities approach focuses on what people are able to do and be”.

Access does not Guarantee Autonomy

Critical Insights/Emerging Arguements

Emerging Argument
Although the neuro-headset provides access to advanced interaction systems, most users do not necessarily gain complete control. The only control they gain is control over the device however, control over how their cognitive data is collected, processed or used is something they do not have.

Therefore, this means participation within the system does not guarantee transparency between the owners and their users, especially in practice.

Key Insight
Ownership exists in theory; however, control exists in practice.

Why This Matters
This matters because it has been exposed that access alone is not enough. Especially, if users remain disconnected from the systems that control their data and participation.

Boylston, S. (2019), Chapter 8.
Access ≠ Capability
“Access does not guarantee opportunity”.

Power is Hidden within the Infrastructure

Critical Insights/Emerging Arguements

Emerging Argument
The neuro-headset appears to empower users by enhancing capability through innovative interaction and accessibility. However, deeper analysis revealed that power is concentrated within the higher actors like the companies (4Design, Emotiv) and developers (Software, Creative professionals, Neurable) rather than equally distributed among the participants.

Companies that control the platforms, data systems and infrastructure retain the greatest authority, while the users/consumers generate valuable cognitive data without any equivalent control over how it is used.

Key Insight
The system appears empowering but at the same time, conceals unequal power structures.

Why This Matters
Reflecting on responsibility through a power lens, exposed aspects of the system that would not be immediately visible when you only focus on functionality or innovation.

Responsibility Flows away from Power

Critical Insights/Emerging Arguements

Emerging Argument
Responsibility within the neuro-headset system is unevenly distributed and frequently shifted, deferred or displaced across multiple actors including companies, designers, developers, users and regulators.

Although corporations retain significant control over data systems and the infrastructure, accountability is often moved toward those with the least power within the system because they have no control or say in anything.

Key Insight
Responsibility flows away from those with power.

Why This Matters
This matters because responsibility gets difficult to assign due to the various actors and timeframes affected by control, impact and accountability operating within the system.

Transition Design/Systems Design literature: Distributed responsibilities
“Responsibility is shared across systems”.

The System Externalises Long-Term Consequences

Critical Insights/Emerging Arguements

Emerging Argument
The neuro-headset creates immediate value through its innovation, accessibility and cognitive interaction. Most of the ethical, social and environmental consequences are displaced across time (NOW & FUTURE), geography (foreign countries outside of research) and less powerful groups (patients, children, users with motor neurone issues).

The benefits are experienced in the present, while long-term risks surrounding privacy, surveillance, environmental impact and cognitive autonomy gradually creep up to the surface; these remain uncertain and difficult to trace due to regulations lagging and not being able to keep up with technological advancements.

Key Insight
Benefits are immediate whereas consequences are deferred.

Why This Matters
This matters because my understanding of responsibility has been expanded. Leading me to consider the potential implications of the wider life-cycle and future of society impacts.

McDonough, W. and Braungart, M. (2002) *Cradle to Cradle Design: Remaking the Way We Make Things*
Externalities
“Costs are often hidden or displaced”.

Irwin, T. (2015) Transition Design
Life-Centred Design
“Design must consider impacts on all living systems, not only humans”.

Emerging Position

Critical Insights/Emerging Arguements

The Emotiv EPOC neuro-headset **expands cognitive capability and accessibility**. However, behind all of this is a system that **redistributes power unevenly, obscures responsibility and externalises long-term ethical consequences**.

Capability ≠ Autonomy

Access ≠ Control

Innovation ≠ Responsibility

Climate, Social and Racial Justice Principles

Throughout this project, my investigation into the **Emotiv EPOC neuro-headset** has unexpectedly become increasingly connected to wider questions that surround **responsibility, justice, accessibility and long-term technological impact to society and the environment**. The **UAL Climate, Racial and Social Justice Principles** have helped frame my enquiry beyong the project itself. Leading to the encouragement in allowing me to think critically in depth about **who benefits, who is excluded from technological systems and how every design decision made has the potential to shape possible futures.**

Principle 03 – Futures Thinking

Climate, Social and Racial Justice Principles

Reflection

Principle 03 supported me when it came to thinking beyond the **functionality of the neuro-headset** alone and instead, focus on the **wider futures that systems like this contribute to producing**. The neuro-headset technology presents opportunities for **accessibility, interaction and innovation** but it also raises long-term questions that surround **cognitive privacy, surveillance dependency and data ownership issues**.

Through the responsibility map I created, the system revealed how many **consequences are hidden and displaced into the actors with less power and the future**. Particularly, around **regulation, cognitive autonomy and environmental impact,** when I considered these ideas, my project was shifted from analysing a product to critically examining the **future systems and power structures** connected to a brain-computer interactive technology.

Key Insight
Innovation does not only shape present interaction but expands into future social and ethical conditions.

Principle 04 – Human Equity and Justice

Climate, Social and Racial Justice Principles

Reflection

Principle 04 became a central part of my investigation into **capability, access and power** within the neuro-headset system and even though the technology seems empowering due to the amount of capability it enhances, deeper analysis has revealed that **participation within the system is unevenly distributed** as I have stated throughout the process book.

It is evident that some users **gain increased capability and interaction**, while others also experience **dependency issues, limited control or even total exclusion over how their cognitive data is collected and used.** Making me question who truly benefits from systems like this, I wonder whether technological access alone can be made **equitable if the actors who use the product did not retain such meaningful autonomy or control.**

The project has challenged me to think critically, **through system, capability, responsibility and trade-off maps because without those maps in aid of my critical analysis and thinking**, I would have prioritised convenience and innovation while overlooking **hidden inequalities embedded within systems and infrastructures.**

Key Insight
Access does not automatically create equity, autonomy or agency (the capacity to act).

Connection to my Enquiry

Climate, Social and Racial Justice Principles

Reflection

Both principles are deeply connected throughout the development of my enquiry. Principle 04 felt relevant because my project **consistently focuses on the uneven distribution of responsibility, access and cognitive autonomy within technological systems.**

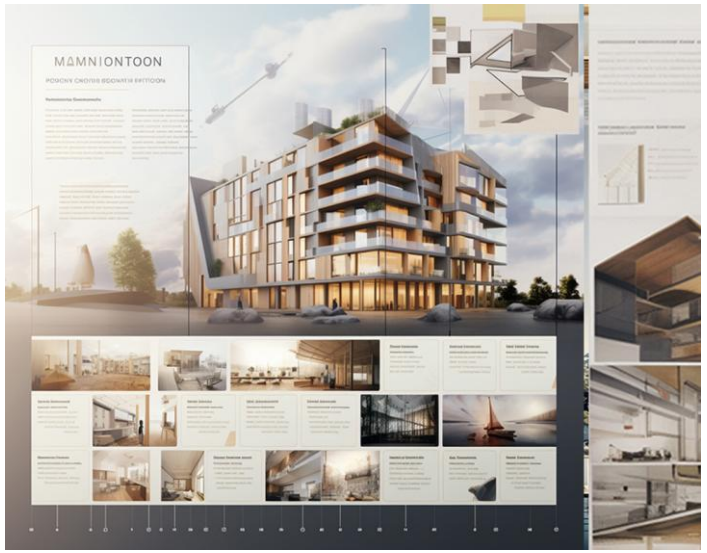
On the other hand, principle 03 became increasingly important as I evolved my project. Through the deep analysis of the neuro-headset, it was revealed to me that many **ethical and environmental consequences extend far beyond immediate interaction.** Therefore, it raised questions about the kind of **future the wider society** may potentially live through now that neurotechnology is **rapidly emerging and getting normalised.**

Together, these principles have helped me understand responsible UX design as something that **extends beyond usability and functionality.** It involves the questioning of who benefits, who bears the hidden costs, how design decisions shape future relationships between people, technology and power within the system.

Responsible UX design extends beyond usability and functionality.

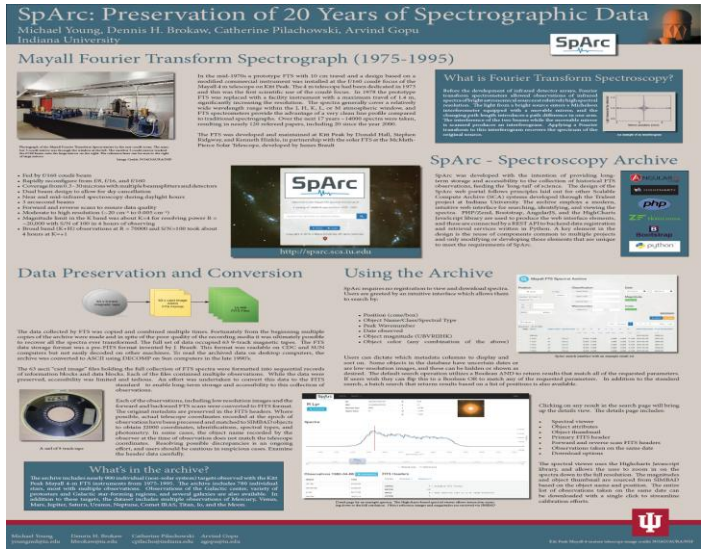
Visual References

Visual Artefact Development



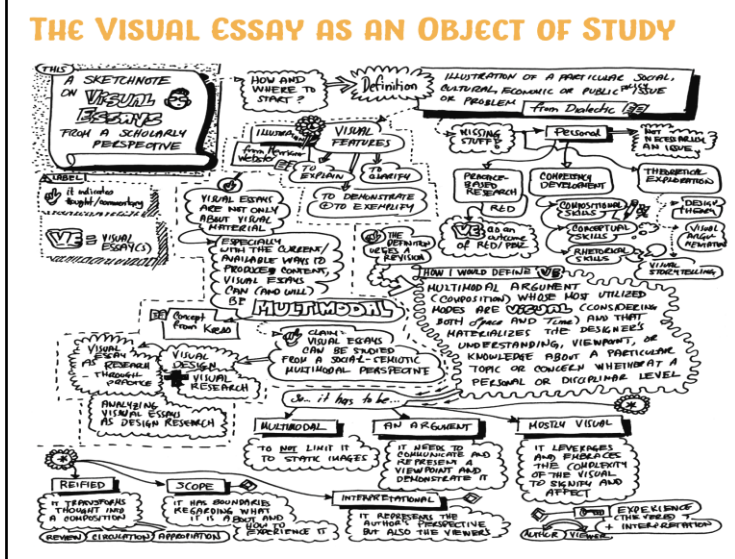
Critical architecture board

I looked at this due to its visually heavy and layered presentation style. What interested me was that the board was able to communicate a strong critical position through visual hierarchy before the viewer began reading the text. Making me realise; my own visual artefact should rely more on composition, contrast, imagery and structure to communicate meaning rather than being overloaded with written information.



Systems critique poster

I looked at this poster because it takes an almost opposite approach in terms of visual hierarchy and structure. Unlike the architecture board, this one relies heavily on text and written analysis to communicate. Making realise although visuals are important to incorporate, my visual artefact is not simply a poster but a visual essay and because of this, the artefact needs to balance a strong visual communication between both text and imagery.



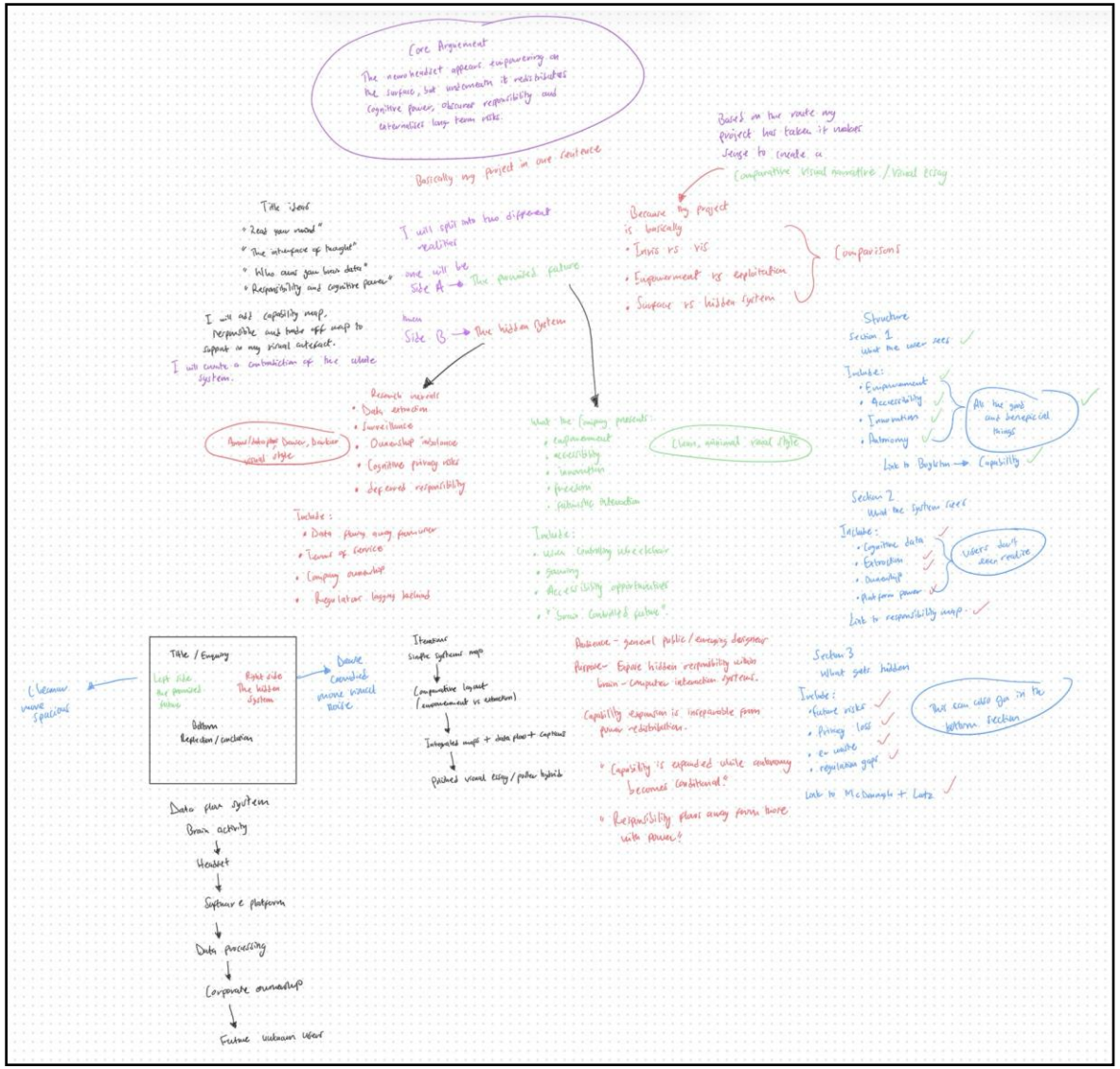
Visual essay

I looked at the this it successfully combines the strongest aspects of the previous two being the theoretical and visual communication. I believe this method would work best because it allows me to communicate both the visible and invisible aspects of the neuro-headset system. Presenting ideas of capability, access and power on the surface while exposing hidden power imbalances and unfair responsibility distributions.

Composition of Ideas and Colour Decisions

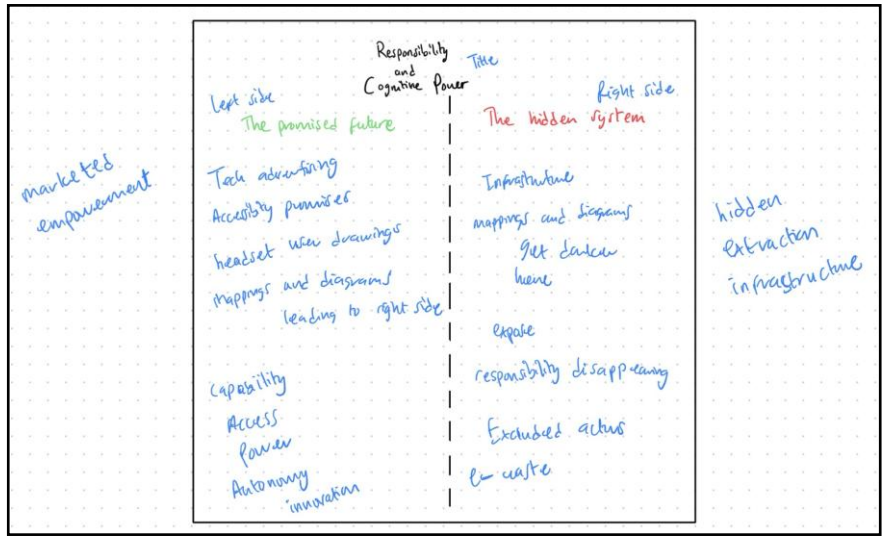
Visual Artefact Development

- Created an **initial brainstorm** to support me in the **generation of ideas** for my visual artefact.
- **My core argument: The neuroheadset appears empowering on the surface, but underneath it redistributes cognitive power, obscures responsibility and externalises long-term risks.**
- I had to find a way to visually display that which led me to something like a **split visual essay**.
- I included a brainstorm of **different titles** to use.
- A rough **self discussion** on what my artefact should include and how it should be structured **(I will have various iterations displayed on the next slides).**
- Deciding what **colours** to use and where.
- My aim is to present everything I have talked about on my process book onto my visual artefact in a more **clarified and educated display.**

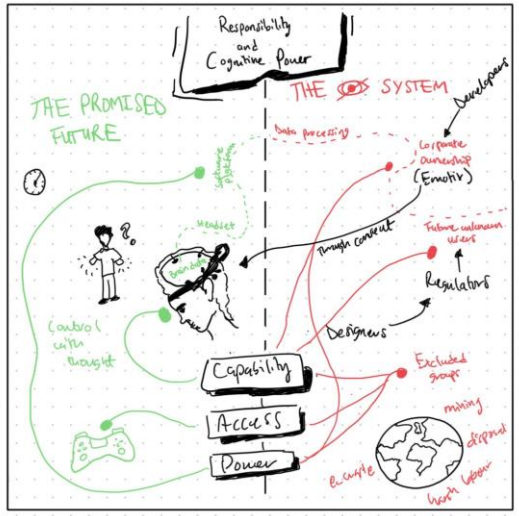


Iteration Sketches

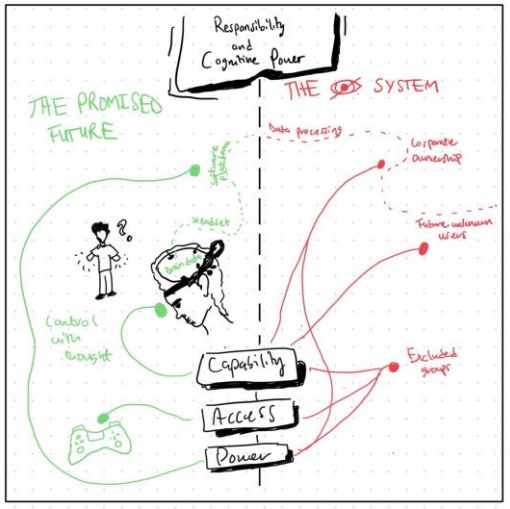
Visual Artefact Development



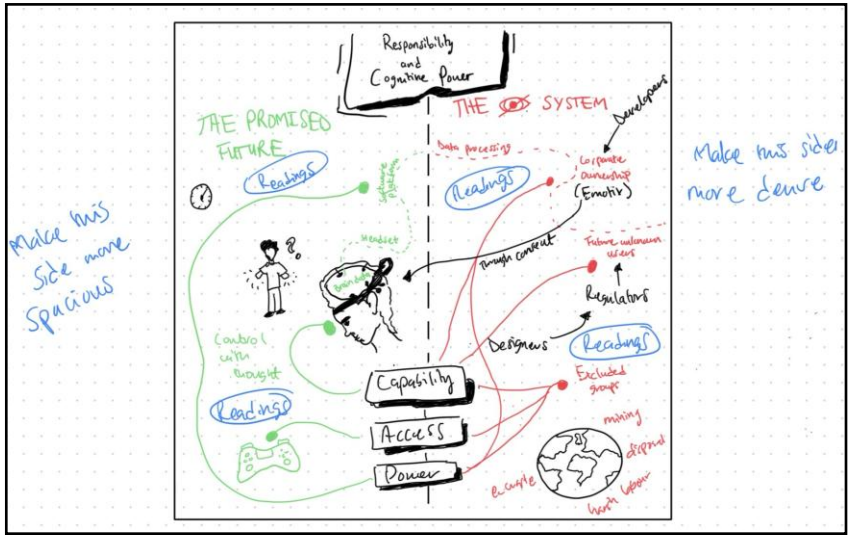
1. I made a quick split composition concept just to help imagine the idea my visual artefact would be going for. Adding notes around and within the piece informing and reminding me what would need to go where and how.



3. I then added the responsibility distribution into my artefact just to get an idea of how it would look in there. I am still not sure on how I will be able to put everything together and keep a clean looking visual artefact. However, this is the reason for iterations, planning to get me ready for the main task.



2. This was when I started considering a certain design approach, a way I could present my accumulated information yet still display a visually appealing artefact. Using lines, illustrations and annotations to communicate, I introduced capability, access, power within a data flow.



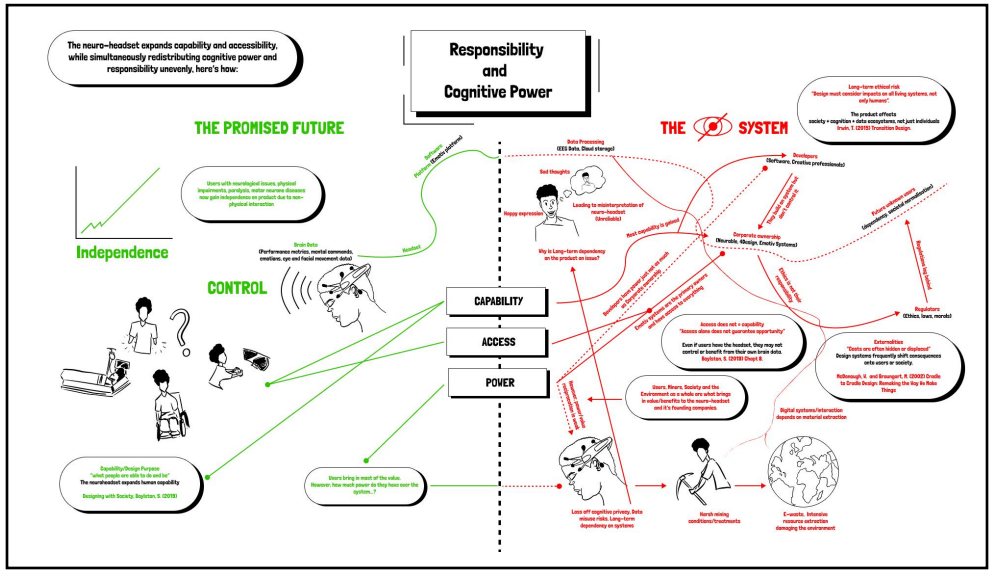
4. A few more annotations were made. Another which was important being the annotations I made on the outside of my iteration plan. On the left side stating; “make this side more spacious” and the right side stating; “make this side denser”. My aim is to convey and display contrast. Which I want to do without having to say but show instead.

Final Artefact Direction

For my visual artefact, I aimed for **a visual comparative essay**. Paying attention to **visual hierarchy, argument, contrast, systems thinking, visual flow with narrative flow.** Most importantly, I created a **split reality composition** because I wanted the **structure itself to be making the argument** and have **contrast communicate** before someone even reads.

This would be creating **a LEFT side vs RIGHT side** structure, which I feel I produced successfully. The left side **feeling optimistic, accessible, breathable, empowering and simple**. However, the right side feels **infrastructural, dense, overwhelming, hidden and imbalanced**. This was the goal for my visual hierarchy.

At the **top left corner** of my artefact, there is a statement I made. This statement was made to give the viewers an idea of what I **aim to showcase/communicate** through my visual artefact but also help **unify my entire piece as one argument.**



How did I fit the maps in you may ask? **The capability, responsibility, trade-off/externalities were merged into my map.** These **became flows, annotations, consequences and system movement**.

I included, each theme and subject idea being**; capability, access, power, responsibility, externalities, futures thinking and systems thinkings**. Integrating them in my visual artefact, allowing them to **operate together**. Reflecting heavily on my **enquiry statement and question (Pg 18).**

Following my visual artefact carefully, you notice that not only is **everything connected with lines**, but **transitions are present**. For example; from a **strong solid line to a broken-down dashed line to highlight collapse, instability or displacement in the system**. Additionally, to show **subtle/even** more hidden connections I used **faded lines** to present them.

I made sure to use **my own illustrations** instead of stock-images or realistic renders because it would have made it appear more promotional or corporate instead of **analytical and critical.**

Embedding the **theoretical frameworks** worked as **supporting critical anchors**. Making it feel integrated and not forced into my piece.

The centre holds**; Capability, Access and Power**. These act **as analytical lenses** that **connect both realities** together. You can see annotation lines coming from each lens. Subsequently, **solid lines from the left side that cross into the right side fracture into dashed lines**.

The neuro-headset expands capability and accessibility, while simultaneously redistributing cognitive power and responsibility unevenly, here's how:

Responsibility and Cognitive Power

THE PROMISED FUTURE



Independence

Users with neurological issues, physical impairments, paralysis, motor neurone diseases now gain independence on product due to non-physical interaction

CONTROL



Brain Data
(Performance metrics, mental commands, emotions, eye and facial movement data)



Software Platform (Emotiv platform)

Headset

CAPABILITY

ACCESS

POWER

Capability/Design Purpose
"what people are able to do and be"
The neuroheadset expands human capability

Designing with Society, Boylston, S. (2019)

Users bring in most of the value. However, how much power do they have over the system...?

THE SYSTEM

Long-term ethical risk
"Design must consider impacts on all living systems, not only humans".

The product affects society + cognition + data ecosystems, not just individuals
Irwin, T. (2015) Transition Design.

Data Processing
(EEG Data, Cloud storage)

Sad thoughts



Happy expression

Leading to misinterpretation of neuro-headset (Unreliable)

Why is Long-term dependency on the product an issue?

Most capability is gained

Developers
(Software, Creative professionals)

They build on system but don't control it

Corporate ownership
(Neurable, 4Design, Emotiv Systems)

Future unknown users
(dependency, societal normalisation)

Regulations lag behind

Regulators
(Ethics, laws, morals)

Developers have power just not as much as Corporate ownership

Emotiv systems are the primary owners and have access to everything

Ethics is not their responsibility

Access does not = capability
"Access alone does not guarantee opportunity"

Even if users have the headset, they may not control or benefit from their own brain data.
Boylston, S. (2019) Chapt 8.

Externalities
"Costs are often hidden or displaced"
Design systems frequently shift consequences onto users or society.

McDonough, W. and Braungart, M. (2002) Cradle to Cradle Design: Remaking the Way We Make Things

However, power/value reciprocation is weak

Users, Miners, Society and the Environment as a whole are what brings in value/benefits to the neuro-headset and it's founding companies.

Digital systems/interaction depends on material extraction



Loss off cognitive privacy, Data misuse risks, Long-term dependency on systems



Harsh mining conditions/treatments



E-waste, Intensive resource extraction damaging the environment

Responsibility, Cognitive Power and Uneven Capability within the Emotiv EPOC Neuroheadset System

Within this project unit, responsibility was investigated within contemporary design systems. Using my chosen artefact, the Emotiv EPOC Neuroheadset from the Designer Maker User exhibit, I examined how responsibility is formed, shared, deferred and obscured within complex systems and how these decisions affect people, communities and ecosystems.

My chosen artefact, The Emotiv EPOC Neuroheadset (2007), was designed by Emotiv Systems and 4Design. It is a brain-computer interface (BCI) that allows users to interact with digital systems using brain signals. It does this by detecting electrical activity within the brain and translating these signals into commands capable of controlling technological devices. The neuroheadset presents a wide range of possibilities for accessibility, particularly for users with paralysis, motor neurone diseases and physical impairments. However, although the neuroheadset appears empowering through cognitive interaction and accessibility, deeper analysis revealed that responsibility, power and control are unevenly distributed throughout the wider system.

To communicate this contrast, I created a comparative visual essay exposing both the visible promises and hidden consequences of the neuroheadset. Through this investigation, my understanding of UX design expanded from just usability and interaction alone towards wider questions that revolve around responsibility, autonomy, power and long-term systemic impact.

At the beginning of the project, I was drawn to the neuroheadset and that was because of how different it felt compared to the other artefacts within the exhibition. It did not feel like a standalone product but instead a system with potentially serious implications. This connected directly to systems thinking and Boylston's idea that contemporary challenges are interconnected and cannot be understood in isolation (Boylston, S. 2019, Section 1/Chapt. 1). The neuroheadset exists within a wider ecosystem that involves cognitive data extraction, cloud infrastructure, platform ownership, regulation, manufacturing and long-term societal impact.

Initially, the neuroheadset excited me because it felt like a genuine step into the future. The idea of controlling systems through thought alone seemed empowering, especially for users with disabilities and severe physical impairments. However, this excitement quickly became mixed with concerns surrounding cognitive privacy, behavioural data, surveillance and hidden infrastructure. The project therefore shifted from analysing a product towards analysing the wider system operating behind it.

Through secondary research and systems mapping, I identified three analytical lenses that became central throughout my investigation: capability, access and power.

Capability became the first major lens within my analysis. Boylston states that “the capabilities approach focuses on what people are able to do and be” (Boylston, S. 2019, Designing with Society, Introduction). This directly connected to the neuroheadset because the device expands functional capability by enabling interaction without physical movement. Users who previously depended on caretakers or physical interfaces may gain increased independence through cognitive interaction.

However, deeper analysis revealed that this capability is conditional. The neuroheadset depends on continuous cognitive data extraction including emotional responses, behavioural metrics and mental commands. Although users gain functionality, they also become increasingly dependent on the system itself. This led to one of my key arguments:

Capability expansion depends on cognitive data extraction.

The headset therefore redistributes capability unevenly. While some users gain increased accessibility and interaction, others experience new vulnerabilities, dependency and reduced cognitive autonomy.

This directly challenged my earlier assumptions about assistive technology. Before the project, I viewed accessibility technologies primarily as beneficial tools. However, I began recognising that expanded functionality does not automatically create autonomy.

This connected to my second major argument:

Access does not guarantee autonomy.

Although users gain access to the neuroheadset, they do not necessarily gain meaningful control over how their cognitive data is collected, processed or used. Boylston argues that “access alone does not guarantee opportunity” (Boylston, S. 2019, Chapt. 8). This became highly relevant within the neuroheadset system because participation does not equal transparency or agency.

Research into Emotiv’s terms and conditions revealed that although users technically retain ownership of their data, the company retains significant practical control through its software and infrastructure (Emotivcloud.com, 2025, Emotiv Terms of Conditions & Privacy Policy). Users grant the company extensive rights to analyse and reproduce cognitive data collected through the headset.

This created another key insight within my project:

Ownership exists in theory, while control exists in practice.

As my investigation developed, I increasingly questioned whether digital participation could ever be considered equitable if users remain disconnected from the systems controlling their own data.

The third analytical lens focused on power. Looking at the system through a power lens exposed aspects of the neuroheadset that were not immediately visible when focusing only on innovation or usability. Although the device appears empowering on the surface, power is concentrated within the system infrastructure.

Companies such as Emotiv retain significant authority through ownership of cloud systems, EEG data infrastructure and platform control (Emotiv.com, 2026, Wearable EEG Sensor Headset: A Buyer's Guide). Meanwhile, users generate the valuable cognitive data without equivalent control over how it is used.

This revealed another major argument:

The system appears empowering while concealing unequal power structures.

This shifted my understanding of UX design away from improving interaction alone towards questioning who controls systems, who benefits from them and who absorbs the risks.

To deepen this critique, I produced a responsibility map and a trade-off/externalities map. These maps helped visualise how responsibility operates across multiple actors, timeframes and systems.

The responsibility map revealed that responsibility frequently flows away from those with the most power. Designers and companies often claim responsibility for innovation, accessibility and usability because these outcomes benefit their reputation and commercial success. However, responsibility for ethical consequences, cognitive privacy and long-term risks is frequently deferred towards users or regulators.

This creates a significant accountability gap within the system. Responsibility becomes difficult to assign because control, impact and accountability are distributed across multiple actors and timeframes.

At the same time, my trade-off map exposed how the neuroheadset externalises long-term consequences across geography, time and less powerful groups.

Although the system appears highly digital, it remains dependent on physical extraction, manufacturing and environmental cost. Intensive mining and material sourcing are required to produce batteries, sensors, PCB boards and electrodes. At the end of the product lifecycle, improper disposal contributes towards electronic waste and environmental damage (CarbonClick, 2025).

This strongly connected to McDonough and Braungart's argument that "costs are often hidden or displaced" (McDonough, W. and Braungart, M. 2002, Cradle to Cradle Design: Remaking the Way We Make Things).

One of the most important realisations throughout my project was understanding that digital systems are still material systems. Although the neuroheadset appears futuristic and immaterial on the surface, it remains connected to extraction, labour, manufacturing and environmental degradation.

This significantly expanded my understanding of responsibility within UX design. Instead of focusing only on immediate interaction, I began considering wider ecological and societal futures created through technological systems.

These ideas directly influenced the development of my final visual artefact.

Throughout the project, I explored multiple visual references including critical architecture boards, systems critique posters and visual essays. Comparing these approaches helped me understand that my artefact needed to balance theoretical information with visual communication.

This led me towards creating a comparative visual essay.

My artefact visually contrasts two realities operating within the neuroheadset system. The left side, titled “The Promised Future,” communicates accessibility, empowerment, capability and innovation. The visual hierarchy intentionally feels spacious, optimistic and breathable.

In contrast, the right side, titled “The System,” exposes hidden infrastructure, data extraction, environmental consequences, dependency and unequal power structures. This side feels denser, more overwhelming and visually fragmented.

The structure itself therefore becomes part of the argument.

Rather than inserting my maps separately, I integrated them directly into the visual artefact through flows, annotations, system movements and fragmented transitions. Solid lines gradually fracture into dashed lines to represent instability, collapse and displacement within the system.

Capability, access and power were placed centrally within the composition as analytical lenses connecting both sides together. I also embedded theoretical references throughout the artefact as supporting critical anchors. Additionally, I intentionally illustrated the imagery myself instead of using stock photography because I wanted the piece to feel analytical and critical rather than corporate or promotional.

Reflecting on the project overall, my understanding of UX design changed significantly. Initially, I viewed UX primarily through interaction, aesthetics and usability. However, this project revealed that UX design also shapes systems affecting autonomy, participation, power and long-term futures.

The project challenged me to think critically beyond surface-level interaction and consider the wider ethical, environmental and social consequences embedded within emerging technologies.

Ultimately, the Emotiv EPOC neuroheadset expands accessibility and cognitive interaction while simultaneously redistributing power, responsibility and risk unevenly throughout the system. Although the technology promises empowerment, deeper systems analysis reveals hidden dependencies, unequal control structures and displaced consequences.

This project fundamentally changed how I view responsible design. I now understand UX design not simply as designing interfaces, but as shaping relationships between people, systems, power and future conditions.

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